

Aryaman Gupta

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RESEARCH OBJECTIVE

My research aims to make robot foundation models (RFMs) safe and reliable by turning their failures/anomalies into a learning signal. I develop methods that discover interpretable failure modes from deployment data, monitor reasoning-action misalignments at runtime, and steer them toward safe, successful task execution at test time with minimal human supervision. Going forward, I aim to build safety-reasoning models that generalize across tasks and embodiments, enabling RFMs to monitor and correct their behavior, and explore post-training RFMs with reinforcement learning using dense failure signals.

EDUCATION

- **Stanford University** Stanford, CA
Ph.D. in Aeronautics and Astronautics; Advisor: Somil Bansal Jan 2025 – Present
- **Stanford University** Stanford, CA
M.S. in Aeronautics and Astronautics; Advisor: Somil Bansal Jan 2025 – Dec 2026 (Exp.)
- **Indian Institute of Technology (BHU)** Varanasi, India
B.Tech. in Electronics Engineering; Advisor: Om Jee Pandey Nov 2020 – May 2024

PUBLICATIONS

- [1] **Aryaman Gupta***, Yusuf Umut Ciftci*, Somil Bansal, “Unsupervised Discovery of Failure Taxonomies from Deployment Logs”, *IROS’26*
- [2] **Aryaman Gupta***, Kaustav Chakraborty*, Somil Bansal, “Detecting and Mitigating System-Level Anomalies of Vision-Based Controllers”, *ICRA ’24*
- [3] A.M. Ali, **Aryaman Gupta**, H.A. Hashim, “Deep Reinforcement Learning for Sim-to-Real Policy Transfer of VTOL-UAVs Offshore Docking Operations”, *Applied Soft Computing Journal*
- [4] Neha Sharma, **Aryaman Gupta**, Sivala Deepak, Om Jee Pandey, “Node Fault Prediction Assisted Small-World IoT Networks Using ML Frameworks”, *IEEE ANTS’24* [Best Paper Award]

*Equal Contribution

PREPRINTS

- [1] **Aryaman Gupta**, Xiaoyue Wu, Rana Soltani Zarrin, Mark Van der Merwe, “GUARD: Anomaly-Informed Q-Steering for Generative Robot Policies in Dexterous Manipulation”, *arXiv ’26*
- [2] Sathwik Karnik, Joseph Lee, **Aryaman Gupta**, Somil Bansal, “When Reasoning Helps Action: Monitoring and Steering Chain-of-Thought in Vision-Language-Action Policies”, *arXiv ’26*
- [3] Kaustav Chakraborty, **Aryaman Gupta**, Somil Bansal, “Enhancing Safety and Robustness of Vision-Based Controllers via Reachability Analysis”, *arXiv ’24*

EXPERIENCE

- **Safe and Intelligent Autonomy Lab** Stanford, CA
Graduate Student | Prof. Somil Bansal Aug 2024 - Present
 - **Research Area:** Policy Steering, Runtime Failure Monitoring, Failure-Guided Policy Improvement.
 - Designed test-time monitors that flag VLA reasoning-action misalignment and steer towards visually accurate chain-of-thought and actions, improving its performance, reliability, and human interpretability.
 - Developed an unsupervised multimodal-LLM pipeline that discovers interpretable failure taxonomies from deployment logs, enabling targeted policy improvement and runtime failure monitoring.
 - Designed Q-guided safety filtering for high-speed systems, such as F1-Tenth cars and drones operating under visuomotor policies, solely through perception inputs without privileged information.

- Honda Research Institute - USA**

Research Internship

 - **Research Area:** Anomaly-informed test-time steering of generative policies for dexterous manipulation.
 - Proposed GUARD, which learns a Q-function from sparse task rewards and dense uncertainty scores, and uses its gradients to steer a frozen flow-matching policy toward successful, in-distribution actions.
 - Designed an offline-to-online Q-learning scheme that adapts the Q-function to the steered policy.
 - Achieved 96%/80% average success on four Adroit tasks under nominal/perturbed dynamics, outperforming the strongest baseline by 21/20 pp while keeping the lowest uncertainty score.
 - Demonstrated zero-shot sim-to-real transfer on in-hand screwdriver rotation and reorientation tasks.

San Jose, CA

June 2026 - Sept 2026
- University of Southern California**

Research Internship | Prof. Somil Bansal

 - **Goal:** Provide safety guarantees on vision-based controllers leveraging knowledge of their failures.
 - Mined failure modes of an aircraft taxiing system by computing Neural Reachable Tubes (NRTs).
 - Developed a runtime vision-based failure monitor and fallback mechanisms, reducing failures by 42%.
 - Fine-tuned policy on augmented failure dataset, reducing failures by 20% and prediction error by 10%.
 - Applied Conformal Prediction for probabilistic safety guarantees enabling deployment under uncertainty.

Los Angeles, CA

May 2023 - Aug 2024
- Carleton University**

Research Internship | Prof. Hashim Mohamed

 - **Goal:** Develop docking mechanism for VTOL-UAVs on offshore charging platforms using Deep-RL.
 - Built a custom environment for UAV landing with JONSWAP model-based hydrodynamic disturbances (waves) acting on the docking station, making it oscillate on the water surface.
 - Implemented DQN and PPO algorithms to compare performance among value and policy-based agents.
 - Agents successfully learnt to land precisely and safely through a carefully designed reward function.

Ottawa, Canada

Jan 2023 - Dec 2023
- Indian Institute of Science**

Research Internship | Prof. Bharadwaj Amrutur

 - **Goal:** Develop centralized multi-agent exploration and vision-aided dynamic collision avoidance systems.
 - Implemented RRT-exploration algorithm and map merging on a multi-robot fleet for collective SLAM.
 - Performed real-time map updates, detecting obstacles with 3D Object Detection through infrastructure cameras and using Gradient Descent to optimize obstacles' 3D real-world position for avoidance.
 - Developed ROS packages that can be configured and implemented on custom hardware testbeds.

Bengaluru, India

May 2022 - July 2022
- Changwon National University**

Research Internship | Prof. Oh-Seol Kwon

 - **Goal:** Develop a deep learning architecture for efficient object detection in low-resolution aerial images.
 - Combined Faster R-CNN, Edge Enhanced Network, and SRGAN architectures for the joint task.
 - Performed End-to-End Training of the entire pipeline, feeding detector's loss to SRGAN network.
 - Obtained testing accuracies of 95.5% on COWC and 83.2% on OGST datasets.

Changwon, South Korea

Mar 2022 - July 2022

SELECTED ACHIEVEMENTS

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|--|------|
| • Qualcomm Innovation Fellowship – among 16 selected teams out of 400+ applicants | 2026 |
| • Annenberg Scholarship, University of Southern California | 2024 |
| • Best Paper Award – IEEE Advanced Networks and Telecommunication Systems | 2024 |
| • Best Undergraduate Thesis – among 146 students in ECE IIT BHU | 2024 |
| • IUSSTF-Viterbi Scholarship – among top 15 students selected across India | 2023 |
| • DAAD-WISE Scholarship | 2023 |
| • Mitacs GRI Scholarship | 2023 |
| • Felicitation by IIT BHU for technical contributions, leadership, and mentorship to fellow students | 2023 |
| • 2nd Position – across India in All IITs Robotics Association Challenge | 2021 |

RELEVANT COURSEWORK

- **Graduate Courses**

- CS-224N (Natural Language Processing)
- AA-276 (Principles of Safety-Critical Autonomy)
- AA-228V (Validation of Safety-Critical Systems)
- ENGR-205 (Introduction to Control Design Techniques)
- CS-224V (Agentic AI)
- AA-228 (Decision Making Under Uncertainty)
- AA-203 (Optimal and Learning-Based Control)
- AA-242A (Classical Dynamics)

- **Undergraduate Courses**

- MA-101 (Real Analysis)
- MA-102 (Linear Algebra)
- MA-202 (Probability and Statistics)
- CSO-102 (Data Structures and Algorithms)
- CSO-332 (Ubiquitous Computing)
- CSO-458 (Soft Computing)

SKILLS AND INTERESTS

- **Languages and Libraries:** Python, C++, MATLAB, Julia, JAX, Hugging Face, PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, OpenAI Gym, X-Plane, CARLA, MuJoCo, mjlal
- **Technologies and Tools:** ROS, ROS2, Flightmare, AutowareAI, Open3D, Linux, Git, Docker, L^AT_EX

COMMUNITY SERVICES AND INVOLVEMENTS

- Reviewer – CoRL (2025, 2026), IROS (2026), RA-L (2025, 2026), T-RO (2025)
- Working Group Member – Stanford Center for AI Safety
- Technical Lead – RoboReG, a student-led robotics research group at IIT BHU

STUDENT MENTORSHIP

- Joyce Yang (Cornell BS CS), Tito Rosas (UCLA BS ME)